Scholarship and Fellowship Platform: Full Research and Build Plan

Oct 29, 2026 · @Pranshu sahu

1. TL;DR

Build one end-to-end golden path for one scheme, prove a second scheme runs by configuration alone, and make "AI flags, officer decides" visible on every screen. That combination, not feature count, is what a national-level jury remembers. This plan is research-grade: it lists every problem, edge case and small detail worth knowing, then an order to build in.

Six decisions that matter most

1. **Primary scheme:** National Fellowship and Scholarship for ST students. It is central sector, has one clean flow, and its A.Y. 2026-27 cycle is live now.
2. **Second scheme:** Post-Matric ST, as a config file only. This proves the rule engine is generic.
3. **Positioning:** a scheme-pack platform. The tribal pack (ST certificate checks, State ST lists, PVTG and Divyangjan equity cuts, tribal-language notices) is the showcase. The engine itself serves any scholarship (SC, OBC, minority, merit, state schemes).
4. **If only four features:** pre-flight "will this pass?" check, deficiency copilot, Rule Studio with what-if simulator, officer cockpit with audit sampling.
5. **Mock honestly:** Aadhaar eKYC, DigiLocker, PFMS and the NPCI mapper are mock adapters with real response shapes, labelled on screen.
6. **Prove with numbers:** field-level OCR accuracy on 40 to 60 synthetic documents, flag precision and recall on injected faults, and a measured time-to-verify comparison.

Problem statement coverage at a glance

Problem statement phrase	Our module	Section
Configurable scheme rules	Versioned JSON rules + Rule Studio + simulator	5
Application, transparent portal, track status	Applicant portal, pre-flight check, status timeline	6

Problem statement phrase	Our module	Section
Respond to deficiencies, notifications	Deficiency copilot, WhatsApp/SMS/in-app notices	6, 9
OCR and document intelligence	Classify, OCR, extract, cross-check, tamper signals	7
Assist officials to verify faster and more accurately	Officer cockpit, queues, SLA timers, audit sampling	8
Selection	Merit ranking, slots, committee view, explainable rank sheet	9
Post-selection tracking	Payment status, renewal, progress, dropout warning	9
Dashboards, analytics, scheme performance	Admin analytics with equity cuts	10
Speed, transparency, accuracy, accountability, accessibility	One headline metric per goal	10
Final decisions under human oversight	Human gates, reasons, second-officer rule, override tracking	8, 12

Caveat on numbers. Figures in this plan come from search excerpts gathered on 29 Oct 2026, not full documents. Verify any number in its original source before it goes on a slide. Section 16 lists what to verify.

2. Reframing: one engine, scheme packs, tribal first

The problem statement names tribal students, so judges will score tribal fit first. Build the core scheme-agnostic anyway, and ship tribal as the first scheme pack. That gives you a scale story (any scholarship runs on it) without losing the domain depth that wins.

Layer	Generic core (reusable for any scheme)	Tribal pack (the showcase)
Identity	One Time Registration mock, duplicate detection across schemes	Tribe plus State checked against that State's ST list

Layer	Generic core (reusable for any scheme)	Tribal pack (the showcase)
Documents	Income, bank proof, marksheet, admission letter	ST certificate, issuing authority, State-specific formats
Rules	Income ceiling, marks, notified institutes, slots, dates	PVTG flag, Divyangjan slot reservation, home-State rule
Language	Hindi and English, Bhashini for 22 languages	Tribal-language notices (Adi Vaani) with human-vetted templates
Assisted channels	CSC operator mode	Ashram schools, hostel wardens, Eklavya schools, tribal welfare offices
Analytics	Pendency, rejection reasons, payment failures	Equity cuts by PVTG, district, gender, Divyangjan; Scheduled Area districts (verify list)

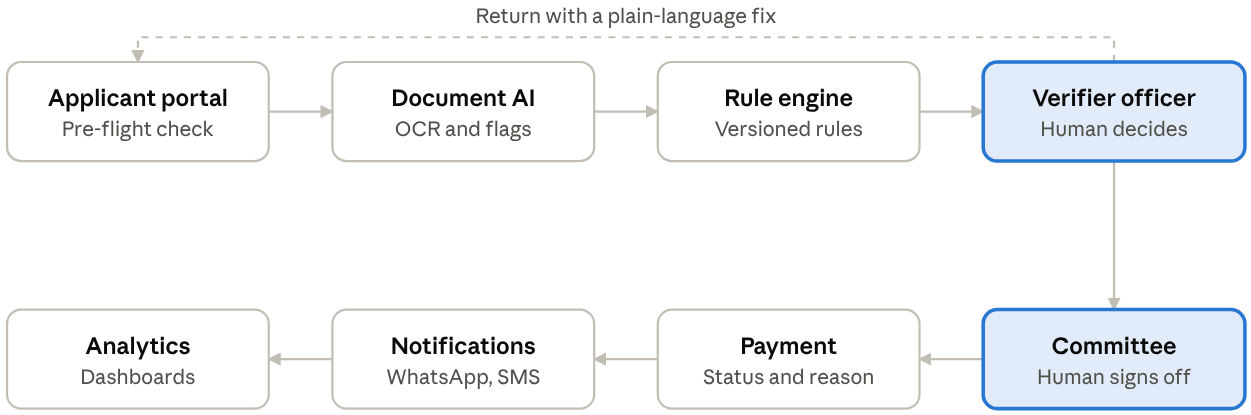
Rules for the framing

- Demo, sample data and language stay tribal-first. A generic demo scores as "another portal".
- Say "one engine, many schemes, we start where exclusion hurts most" once, then show the config proof.
- Do not claim the AI decides tribal identity. States issue and verify certificates; the platform routes doubts to the issuing authority.
- Do not claim you solve all scholarships. Claim the engine accepts them as config.

End-to-end flow with the two human gates

The drawing below shows where AI acts, where humans decide, and where the applicant is sent back with a plain-language fix.

AI prepares the file; two humans decide it



Highlighted boxes are the human decision gates.

AI may flag, pre-fill, rank and explain. Only officers and the committee approve, reject or select.

end-to-end flow · 8 stages, 2 human gates

Read it left to right on the top row, then right to left below. The dashed line sends a flagged file back to the student with a fix, never straight to rejection.

3. Problem map: where today's systems break

Win the first minute with evidence, not adjectives. Each failure below is tied to a feature, so every feature you build answers a documented problem. Figures come from the research pack (audit reports, State notices, news); verify against the original before quoting.

A. Problems with evidence in the research pack

#	Problem	Evidence	Design response
1	Aadhaar-bank seeding blocks payment	Tripura notices with codes like "UID never enabled for DBT"; Arunachal 7,164 students unpaid for seeding reasons; earlier Karnataka report of 4.5 lakh pre-matric students unpaid	Pre-flight seeding check, plain-language fix, payment tracker

#	Problem	Evidence	Design response
2	Slow multi-level verification	Tripura extended Institute, District and State Nodal Officer deadlines for its 2024-25 merit award scheme	Priority queue, SLA timers, pre-checked packets
3	Ghost beneficiaries and fake institutions	CAG Jharkhand: Rs 1.17 cr to 1,482 ghosts from 14 institutes never on NSP. CAG Odisha: 5,185 beneficiaries of 15 ineligible institutions, Rs 15.79 cr. Himachal: Rs 28 cr to nine unaffiliated institutions with reused accounts. CAG minority scholarships (Dec 2025): 8,669 applicants not in institute records	Institute trust score, enrolment cross-check, duplicate account graph
4	Eligible students left out	CAG Tamil Nadu: 35 to 55% of eligible ST students missed in pre-matric and 30 to 42% in post-matric (2019-23); 46% of sampled schools never used pre-matric	Coverage-gap analytics, outreach lists for CSC and schools
5	Duplicate payments	CAG Tamil Nadu: course fees paid twice to 253 ST students	Cross-scheme dedupe on One Time Registration ID
6	Certificate bottleneck	Himachal letter urges linking income, caste and domicile certificates to Aadhaar in DigiLocker so NSP can verify by API; State availability is partial (third party)	DigiLocker pull, OCR fallback, issuer verification request

#	Problem	Evidence	Design response
7	Disputed tribe status	West Bengal and Manipur complaints	AI never decides identity; doubts route to the issuing authority
8	Face-authentication friction	Locked biometrics, poor lighting; CSC fingerprint fallback exists	Assisted mode with recorded consent
9	Rushed approvals (inference, not audit finding)	Slow manual checks push officers to approve without deep scrutiny	Pre-checked packets make careful scrutiny the cheap path

B. Likely problems not in the pack (find a source before claiming)

Problem	Design response
Vague rejections ("documents not proper") with no way to fix	Exact-fix deficiency messages with a guided re-upload
Deadline-day traffic surge	Queue-based pipeline, load test that simulates the last-day spike
Same data re-entered on central, State and institute portals	One verified profile reused across schemes and years
Students never learn they are eligible	Public eligibility checker before registration; coverage-gap outreach
Renewal re-submits every document	Renewal autopilot, no new income certificate where rules allow
Institute nodal officer overloaded and a single point of failure	Delegation, workload balancing, second-officer rules
No traceable grievance path	Grievance ticket with reference ID, SLA and escalation to a different officer
Stipend or fee delays push students to drop out	Payment tracker, dropout early warning

Pitch use. One slide, three sourced numbers, each with its source on the slide: 1,482 ghosts (CAG Jharkhand), 35 to 55% eligible ST students missed (CAG Tamil Nadu), 7,164 unpaid for seeding reasons (Arunachal). Then say what you built against each.

4. Stakeholders and personas

The winning platforms are built around the officer and the operator as much as the student. Ten roles touch a scholarship file; design a screen for each one that you demo, and a permission rule for the rest.

Role	Real context	Main pain	What the platform gives them
Student applicant	First-generation, phone-only, low bandwidth, Hindi or a tribal language	Unclear documents, rejection with no reason, payment failure	Pre-flight check, voice-guided form, exact-fix messages, plain-language payment status
Parent or guardian	Gives consent for under-18 applicants (Pre-Matric covers classes IX and X)	Cannot follow digital forms	Consent step with recorded guardian action, notices in their language
Assisted-mode helper (CSC operator, hostel warden, ashram-school teacher)	Fills forms for many students	Repeats the same errors for every student, fingerprint fallback	Batch queue, consent ledger entry per student, error summary per batch
Institute Nodal Officer	Verifies originals for every student of the institute	Volume near deadlines, no tooling	Risk-ordered queue, pre-checked packets, bulk approve for clean files
District and State Nodal Officers	Second and third verification level for State schemes	Waiting on lower levels, no visibility	Pendency view by institute, SLA timers, escalation

Role	Real context	Main pain	What the platform gives them
Ministry verification officer	Final check for central-sector schemes	Large files, duplicate and fake-institute risk	Decision sheet, duplicate graph, institute trust score
Selection committee	Ranks and signs off	Spreadsheet merit lists, disputed tie-breaks	Explainable rank sheet, waitlist, tie-break audit
Scheme owner or policy maker	Sets rules, reviews performance	Rule changes need code or circulars	Rule Studio with what-if simulator, scheme analytics
Finance and sanction officer	Releases funds	Failed payments, reconciliation	Payment batch file, reason-coded failures, retry list
Auditor or grievance officer	Reviews decisions and complaints	No evidence trail	Read-only audit view, tamper-evident log, appeal path to a different officer

Fictional demo characters (label them fictional on screen): a Gond PhD applicant from a district in Madhya Pradesh, an Institute Nodal Officer at a university, and a ministry officer. Three people carry the whole demo story.

Field research to do before the event (even one conversation changes the pitch):

- One CSC operator: which documents fail most, how long a form takes, what the fingerprint fallback looks like.
- One university scholarship clerk or nodal officer: how verification actually runs, what they check on paper.
- One or two students who applied through NSP: where they got stuck, what rejection messages said.
- Watch the live NSP flow for the current cycle (public pages and FAQs) and screenshot every step so your screens address the real ones.

5. Scheme configuration and rule engine

The rule engine is the heart: every other module reads it. Keep it deterministic and tested, with no AI inside it. A new scheme is a config file plus a Rule Studio session, not a code change.

What a scheme config holds

- Scheme identity, owner department, academic year, version, effective-from and effective-to dates.
- Eligibility rules, each with a stable id, field, operator, value, applies-to (fresh or renewal), severity, and the message shown in each language.
- Required documents per applicant type, with validity rules (for example income certificate fresh-only, issued within the allowed period).
- Form fields and conditional sections.
- Verification chain (who verifies, in what order, with what time limit).
- Selection logic: merit formula, slots, reservations, tie-breaks, waitlist.
- Payment parameters, renewal conditions and communication templates.

```
{
  "scheme": "national-scholarship-st",
  "version": "2025-26",
  "effective": {"from": "2025-07-01", "to": "2026-06-30"},
  "rules": [
    {"id": "income", "field": "family_income", "op": "<=", "value": 600000,
     "applies_to": "fresh", "severity": "hard",
     "msg": {"en": "Family income must be within the ceiling", "hi": "..."}},
    {"id": "institute", "field": "institute_id", "op": "in", "value":
"notified_list_2026", "severity": "hard"},
    {"id": "no_duplicate", "field": "otr", "op": "not_in", "value":
"post_matric_beneficiaries", "severity": "hard"}
  ],
  "documents": ["st_certificate", "income_certificate:fresh_only",
"admission_letter", "bank_proof"],
  "verification_chain": ["institute_nodal", "ministry"],
  "decision_authority": "institute_then_ministry"
}
```

Values above are examples. Verify every value against the current guideline PDF before showing it.

Engine behaviours worth building

- **Three outcomes:** pass, fail, needs-review. Each carries the rule id and the evidence it used. Needs-review is for missing or low-confidence data, never a silent fail.
- **Severity levels:** hard rule (blocks), soft rule (warns), advisory (informs). The officer sees all three.
- **Version pinning:** an application is judged under the rule version in force when it was submitted. A mid-cycle rule change never silently re-judges old files.
- **Maker-checker publishing:** one person drafts a rule change, another approves it. Both are logged.
- **Counterfactual hints:** "Income is Rs 6.4 lakh against a Rs 6 lakh ceiling" is clearer than "ineligible". Also show which other scheme the student may qualify for.
- **Golden tests:** each rule ships with passing and failing test cases that run on every publish.
- **Separation:** eligibility (can apply), verification (documents are valid), selection (ranked within slots) and payment (money released) are four distinct stages with their own rules.

Rule Studio (no-code)

- Form builder, document checklist builder, rule builder, verification chain builder, template editor.
- Version diff and rollback.
- **What-if simulator:** change the income ceiling and see how many synthetic applicants flip from fail to pass, by State and category. This is the most visual proof that the config is real.

Edge cases the rules must express

- Income certificate validity period and financial-year boundaries.
- Fresh versus renewal applicants, and course level (degree, PG, MPhil, PhD).
- Notified institute list changing each year (265 institutes on the 2026 list per the pack; verify).
- Marks as percentage versus CGPA, with a conversion rule per university.
- Age limits, gap years, dual degrees, institute switch mid-year.
- Not holding two overlapping benefits at once.
- Relaxations for Divyangjan, and slot reservations under fellowship guidelines (verify exact terms).

- Attendance or progress conditions for renewal.

Proof that config is enough: two schemes side by side

Config item	National Fellowship and Scholarship (fully built)	Post-Matric ST (config only)
Type	Central sector	Centrally sponsored
Verification chain	Institute or University, then Ministry	Institute, District, State (verify)
Income rule	None for fellowship; ceiling for scholarship	Ceiling per current guideline (verify)
Documents	ST certificate, admission or registration proof, bank proof, plus scheme-specific papers	ST certificate, income certificate, admission proof, bank proof
Selection	Slot-based merit ranking	Entitlement-based approval, no ranking

6. Applicant experience and small details

The applicant journey has ten steps. The big win is catching problems before submission (step 5) and fixing them in one attempt (step 8).

1. **Discover:** a public eligibility checker that needs no login. Five questions in, it says which schemes the student likely qualifies for and which documents to gather. It also lists what to do first if a certificate is missing.
2. **Register:** mobile OTP, then a mocked One Time Registration step (Aadhaar eKYC, face authentication). Show only the last four Aadhaar digits. A fallback path covers failed face authentication: CSC fingerprint in assisted mode.
3. **Fill the form:** mobile-first, Hindi and English, auto-save drafts, progress bar, field-level help, voice guide that reads each question aloud. Prefill from registration and, where available, DigiLocker. Institute search by name or code, IFSC lookup that shows the bank and branch, and a transliteration helper for names.
4. **Upload with a quality gate:** detect blur, glare, cut-off edges, rotation and oversize files on the phone before upload. Guided camera overlay, multi-page capture, compression, resumable upload, and HEIC support.

5. **Pre-flight "will this pass?" check:** a checklist runs before submit and shows each issue with a fix (table below). Warnings never block a student's right to submit; they acknowledge and go on.
6. **Confirm what the machine read:** "We read your family income as Rs X. Is that correct?" Student confirmation turns OCR into a signal the officer can trust.
7. **Submit:** declaration with e-sign (mocked), receipt PDF with application ID and QR code, and a confirmation by SMS or WhatsApp. Allow edit or withdraw until the file is locked for verification.
8. **Track and fix:** status timeline in plain language, with the holding stage, the expected wait from historical medians, and a deadline countdown on any deficiency. Each deficiency is a card with what is wrong, why, a sample photo, and an upload button right there.
9. **Rejection and appeal:** a reason code plus a plain sentence, what the student can do next, and an appeal to a different officer.
10. **After selection:** payment status in plain language, renewal reminders, and a grievance button on every screen.

Pre-flight checks to build

Check	Source of truth	Message the student sees
Bank account seeded for DBT	NPCI mapper (mocked, real reason codes)	"Your account is not linked to Aadhaar for payments. Visit your bank with your Aadhaar; ask for DBT seeding."
Name mismatch across Aadhaar, certificate, marksheet and bank	OCR plus fuzzy matching	"Your name is spelled differently on two documents. Upload a correction affidavit, or confirm which spelling is right."
Income certificate too old or above ceiling	Certificate date and amount vs rule	"This certificate is older than allowed. Get a fresh one from your Tehsil office."
Blurry or cropped document	Image quality model	"The photo is blurry. Place the paper flat in daylight and retake."

Check	Source of truth	Message the student sees
Missing document	Scheme checklist	"You still need your admission letter."
Institute not on the notified list	Institute master data	"Your institute is not on this scheme's list. Here are schemes it is on."
Already a beneficiary of an overlapping scheme	OTR dedupe	"You already receive another scheme's benefit this year."
Bank proof missing the account number or IFSC	OCR fields	"We cannot read the account number. Upload the first page of your passbook."

Small details that separate good from great

- Application and receipt IDs that a CSC operator can read over the phone; QR code on the receipt that opens status.
- Session auto-logout and no cached documents on shared CSC computers.
- Phone-number change and lost-phone recovery without a new registration.
- Name variants: Devanagari and Latin spellings, initials, "Km." or "Kumari" prefixes, surname order, marriage name changes.
- Date format variants (dd/mm/yyyy, Hindi numerals) and caste-certificate number formats that differ by State.
- Under-18 flow with a guardian consent record.
- Disability: UDID capture, accessible UI, scribe or helper flag.
- Text-size toggle, high-contrast mode, screen-reader labels, and a keypad-phone path via SMS keywords.
- Low-data mode: image-light screens, compressed uploads, offline drafts that sync later.
- Document wallet: upload a document once and reuse it across schemes and years, with expiry reminders.
- Quiet hours and a daily cap on notifications, with a delivery receipt per message.

7. Document intelligence

Treat every AI output as a signal with a confidence score and an evidence box, never as a verdict. The pipeline has seven stages, and each one fails safely to a human.

1. **Capture quality gate:** blur, glare, crop, rotation, size, and page count, checked on the device.
2. **Classify:** which document is this (certificate, marksheet, bank proof)? Low confidence goes to the student to confirm.
3. **OCR in Hindi and English,** keeping bounding boxes for every word.
4. **Extract typed fields:** name, number, date, amount, institute, IFSC, tribe, issuing authority.
5. **Validate fields:** format and checksum tests (Aadhaar Verhoeff checksum, IFSC pattern, date sanity, account-number length).
6. **Cross-check:** across documents and against the form.
7. **Score and explain:** pass, needs-review or deficient, with the rule id and the evidence box.

Documents and what to check on each

Document	Key fields	Checks	Common failures
ST certificate	Name, parent name, tribe, State and district, number, issue date, authority, seal or QR	Tribe is in that State's ST list; issuing authority plausible; seal or QR present	Other-State certificate, old handwritten format, name variants
Income certificate	Name, annual income, date, authority	Amount against ceiling; issue date within validity; amount in words equals amount in figures	Expired, ambiguous financial year, words and figures disagree
Marksheet or transcript	Name, roll number, marks or CGPA, course, institute	Marks against threshold; institute and course match the form	CGPA to percentage conversion differs by university

Document	Key fields	Checks	Common failures
Admission or registration letter	Institute, course, session, enrolment number	Institute on the notified list; session is current	Provisional letter, wrong session
Bank proof	Name, account number, IFSC, bank	IFSC valid; name match; account in the student's own name	Joint account, parent's account, passbook cover missing
Aadhaar (masked)	Name, last four digits, DOB	Name and DOB match other documents	Unmasked upload that must be rejected from storage
Disability certificate or UDID	Name, percentage, validity	Name match; validity	Expired
Fellowship papers (verify list)	Registration letter, supervisor certificate, test scorecard	Dates and names consistent	Missing supervisor signature

Cross-checks to implement

Check	How	Output
Name across Aadhaar, certificates, marksheet and bank	Normalise, transliterate Devanagari to Latin, token-set similarity, phonetic match	Match, probable match (show both), or mismatch
DOB and parent's name	Exact and fuzzy match	Flag with both values side by side
Tribe and State against that State's ST list	Lookup table per State	Pass, or route to issuing authority
Income certificate date and amount	Rule engine	Pass, expired, or above ceiling
Institute and course against notified list	Master data lookup by AISHE code	Pass or flag

Check	How	Output
Bank name, IFSC, account reuse	IFSC directory; hash the account number	Valid, invalid, or shared with another applicant
Same Aadhaar or registration ID across applicants or schemes	Hashed identifiers	Duplicate flag with a link to the other file

Tamper signals (report as risk, never as proof)

- File metadata and edit software traces.
- Error-level analysis and compression inconsistencies.
- Font or alignment mismatch within a line, and template deviation against known layouts.
- Missing seal, signature or QR code, or a QR that does not match printed text.
- State plainly in the pitch: a rescanned forgery has innocent metadata, no single method catches a competent forger, and these signals only prioritise human attention.

OCR engine plan

- Primary engine plus a fallback model used for low-confidence fields.
- Candidates: Sarvam Vision or Akshar (Indic documents, coordinate grounding, key-value extraction; accuracy claims are vendor-reported), Qwen3-VL-8B (open, 75.2 chrF++ on a Hindi test in a June 2026 preprint, fits one 24 GB GPU), EasyOCR or Tesseract as a baseline, Indic TrOCR for handwriting, and BharatGen Patram-7B.
- The same preprint found English OCR strength did not predict Hindi accuracy, so test every engine on your own scans.
- Confidence routing (thresholds are starting points to tune): high confidence auto-fills, medium asks the student to confirm, low sends the field to the officer for manual entry.

Rules for using a language model

- Use it to explain flags in plain language and to summarise a file for the officer, with a deterministic template as fallback.
- Extraction must be grounded: if a model-extracted value is not found in the OCR text, discard it.
- Treat document text as data. A document that says "approve this file" must have no effect on the system.

- Never let a model decide eligibility, tribe status or fraud.

Evaluation plan (this is your proof)

- Generate 40 to 60 synthetic documents: six or more fictional State certificate layouts, stamps, handwriting fonts, and phone-photo noise (rotation, shadow, blur).
- Inject twelve known faults: name mismatch, DOB mismatch, expired income certificate, income over ceiling, words versus figures mismatch, wrong tribe for State, missing seal, institute not notified, invalid IFSC, shared bank account, edited text, cropped page.
- Report field-level exact match and character error rate, split by printed versus handwritten and Hindi versus English.
- Report flag precision and recall per fault type, plus time and cost per document.
- Include a slide of failures and what the human fixed. Honesty scores higher than a clean number.

8. Verification and the officer cockpit

The officer decides whether a platform is adopted, so the cockpit is where you spend design effort. Its promise is simple: the careful path becomes the fast path.

Cockpit features

- **Queue ordered by risk and age,** with filters by institute, district, scheme and deficiency type. The risk score shows its components (for example name mismatch, certificate age, duplicate account) so officers can see why a file is high priority.
- **Side-by-side view:** document image with evidence boxes, extracted fields, and the application form, with each AI flag showing rule id, confidence and one plain sentence.
- **Three decisions:** approve, return for correction, reject. Return and reject need a mandatory reason from a coded list plus optional free text.
- **Keyboard shortcuts** for next file, approve, return, open evidence; undo within a short window.
- **Bulk approve for clean files,** always paired with the audit sampling below.
- **Issuer verification request:** when a certificate looks doubtful, send a structured request to the issuing authority and park the file with its own timer. The platform does not rule on tribe status.
- **Duplicate account graph** and **institute trust score** inside the file view.

- **SLA timers,** automatic escalation when a file ages out, delegation when an officer is on leave, and recusal when an officer has a conflict of interest (same family or same institute role).
- **One-page decision sheet** that summarises the file, every signal shown, the decision and the reason, exportable for audit.

Workflow states

State	Who acts	Exits
Draft	Student or helper	Submitted
Submitted and auto-screened	System	Institute verification, or back to student if the pre-flight check failed hard
Institute verification	Institute Nodal Officer	Verified, returned for correction, or escalated
District then State verification (State schemes)	District and State Nodal Officers	Verified or returned
Ministry verification (central schemes)	Ministry officer	Verified or returned
Deficiency	Student	Back to the same stage with a fresh timer
Selection	Committee	Selected, waitlisted, or not selected with reasons
Sanction and payment	Finance officer, then payment rail	Paid, or failed with a reason code
Rejected and appeal	Second officer to reject; different officer for appeal	Final rejection or reinstated
Renewal	Student and officer	Continued or closed

Time limits per state are set in the scheme config, not in code.

Rejection reasons (build a coded list)

Every reason has a code, a plain sentence in each language, a flag for "student can fix this", and a flag for "appealable". Examples: document unreadable, document expired, name mismatch unresolved, income above ceiling, institute not notified, duplicate application, certificate not verified by issuer, bank account not eligible for payment.

Guarding against rubber-stamping

- **Random audit sample:** a fixed share of auto-clean files goes to full manual review. The miss rate becomes a published metric.
- **Blind review on the sample:** for some sampled files the officer sees no AI signals, so you can measure how often AI and human disagree.
- **Override tracking:** log how often officers overrule AI flags and how often AI clean files get rejected on review. A rising override rate triggers a rule or model review.
- **Second-officer rule:** a rejection needs a second officer. Appeals go to a different officer.
- **Evidence-open requirement:** to approve a file with a high-risk flag, the officer must open its evidence first.
- **Bulk-approve guardrails:** only for files with no flags, capped per session, and each batch logged.

Cockpit test to run and report

Have three or four people verify the same 20 synthetic files twice, once with a plain table and once with the cockpit. Report median time per file, error count against known faults, and a one-line quote from each. Even a small test gives you a real speed number for the pitch.

9. Selection, communication and post-selection

These three stages are where most teams stop at a mock screen. Build them as real workflow with logs, because the problem statement names all of them.

Selection

- **Merit ranking from config:** the scoring formula, slot counts, reservations and tie-break order all live in the scheme config. Fellowship slots are about 750 a year (third party; verify).
- **Committee view:** ranked list with each candidate's score components, flags that were resolved, and the verification decision. Any committee change to rank needs a reason and is logged.

- **Waitlist with auto-promotion** when a selected candidate declines, is found ineligible, or fails renewal.
- **Explainable rank sheet:** one page per candidate showing how the score was reached. Export as PDF for the minutes.
- **Tie-break audit:** every tie shows which rule broke it.
- **Equity preview before publishing:** the committee sees the selected pool by State, gender, Divyangjan and PVTG. Surprises are visible before the list goes out.
- **Published list** with application IDs rather than names, an objection window, and a final sign-off step (digital signature mocked).
- Test reservation logic against hand-worked examples. Verify the reservation terms in the current guideline, including slots for Divyangjan.

Communication

Event	Channel	Principle
Registered, submitted	In-app, SMS	Confirm and give the application ID
Deficiency found	In-app, WhatsApp, SMS, voice note	Say exactly what to fix, how, and by when
Deadline approaching	SMS, WhatsApp, helper alert	Count days left; one action
Returned or rejected	In-app, SMS	Reason in plain words, what the student can do, appeal route
Selected or waitlisted	In-app, SMS, WhatsApp	Next steps, documents for sanction
Payment initiated, credited, failed	In-app, SMS, WhatsApp	If failed, the fix (seeding, IFSC, name) and who acts
Renewal due	SMS, WhatsApp, in-app	List only what is needed this year
Grievance update	In-app, SMS	Reference ID, current stage, expected time

Communication rules

- Every message has one action. Keep personal data out of SMS bodies.

- Preferred language per student, with fallback channels and a retry rule. If a notice stays undelivered after a set number of tries, alert the school, hostel or CSC helper.
- India-specific compliance: SMS templates must be registered under TRAI's DLT system and WhatsApp business templates pre-approved. Mock the gateways, but design the templates to fit these constraints.
- **Human-vetted templates for tribal languages.** Adi Vaani covers Santali, Bhili, Mundari and Gondi among others, but an official said Gondi accuracy is low and varies by region. Machine translation may draft; a person approves.
- **WhatsApp status bot:** intents are status, documents needed, deadline, how to fix, and talk to a human. Verify identity by OTP before sharing status. The bot never decides eligibility.

Post-selection tracking

- **Sanction and payment:** sanction order, payment batch file in the format a payment rail expects (mocked, with real reason codes), UTR tracking and reconciliation.
- **Payment status in plain language:** initiated, processed, credited, or failed with a fix. Illustrative reasons to map: Aadhaar not seeded for DBT, account closed or dormant, invalid account number, invalid IFSC, name mismatch, account not in the student's name.
- **Seeding doctor:** turns a failure reason into numbered steps (visit bank, quote these words, come back here to re-verify) and tracks the fix through to re-payment.
- **Renewal autopilot:** pre-fills renewal, skips documents the rules do not need again (for example no new income certificate where renewal does not require one), and asks only for current-year proof.
- **Progress tracking:** semester results, attendance or supervisor progress confirmation for fellows, with reminders to the student and the institute.
- **Dropout early warning:** signals such as missed renewal, payment failures and falling attendance produce a human outreach list. It never penalises a student.
- **Exceptions:** institute change, withdrawal, death or long illness, wrongly paid benefit recovery, stop-payment, and completion or exit.

10. Admin dashboards and analytics

The problem statement lists five goals. Give each one a single headline metric, a definition, and a place on the dashboard, so the jury can see the claim and the measurement side by side.

Goal	Headline metric	Definition	Supporting metrics
Speed	Median and 90th percentile days from submission to verified	Per stage and end to end	Queue age, SLA breaches by institute and district
Transparency	Share of applicants who can see their stage and reason	Applicants with a visible stage plus a reason for any return or rejection	Time to first status, notification delivery rate
Accuracy	First-pass acceptance rate	Files verified with no return	AI flag precision, officer override rate, audit-sample miss rate
Accountability	Share of decisions with a logged officer, time, signals shown and reason	Every decision, no exceptions	Appeal reversal rate, grievance resolution time
Accessibility	Share of applications completed in a regional language or in assisted mode, and completion rate on low bandwidth	By language and channel	Drop-off by step, accessibility test pass rate

Dashboard pages

- **Operations:** pendency by institute, district and officer; SLA breaches; queue age; stuck files with a one-click escalation.
- **Quality:** ranked deficiency reasons, first-pass acceptance, AI flag precision and recall, officer override rate.
- **Fraud radar:** duplicates caught, accounts shared across applicants, institute trust score distribution, files flagged by tamper signals and their outcomes.
- **Equity and coverage gap:** eligible versus applied by district, State, gender, Divyangjan and PVTG. The eligible population comes from census or enrolment data; use synthetic numbers in the demo and say so.
- **Finance:** funds sanctioned, released and utilised; payment failures by reason code; amounts stuck.

- **Grievance:** open tickets by age and stage, repeat causes.
- **Scheme performance:** applications, selected, paid, renewed and dropped out by scheme and year, with a one-line health summary.
- **Public transparency page:** aggregate numbers only, no personal data.

Design details

- Every number has a definition tooltip and a data-freshness stamp.
- Every chart answers a decision ("which institutes need a nudge this week?") and links to the list of files behind it.
- Role-based views: scheme owner, State officer, institute officer.
- Alerts: SLA breach, spike in a deficiency reason, spike in payment failures, rule change impact.
- Avoid ranking officers by speed alone. It rewards rubber-stamping. Show speed together with return rate and audit-sample outcomes.
- Show the effect of your own platform: before and after pendency from your simulated test, labelled as a simulation.

11. Accessibility, language and inclusion

Accessibility is one of the five stated goals, so give it a visible metric and a tested claim. Most teams only translate buttons.

Language

- Hindi and English as the base, Bhashini services for the 22 scheduled languages.
- Adi Vaani (Ministry of Tribal Affairs) translates Hindi and English with Santali, Bhili, Mundari and Gondi; some descriptions add Kui and Garo. An official has said Gondi accuracy is low and varies by region. Use it as a draft tool and keep human-vetted templates for every deficiency, rejection and payment notice.
- Many tribal languages are oral first or have more than one script. Offer audio versions of every key notice rather than text alone.
- Never claim support for a language you have not tested with a speaker.

Low literacy

- Icons next to every step, a read-aloud button on every screen, voice input for free-text answers, and a picture guide for each document.
- Short vernacular explainer clips for the top five errors.

- WhatsApp voice notes from the bot for deficiency and payment messages.

Low connectivity and old phones

- Mobile-first PWA that works on low-end Android devices.
- Compressed, resumable uploads; offline drafts that sync later; text-only fallback.
- SMS keywords for status on keypad phones.
- A camp mode for field staff in remote areas: capture files offline, sync when a signal returns.

Assisted mode

- For CSC operators, hostel wardens, ashram-school and Eklavya school staff, and tribal welfare offices.
- Recorded consent per student; the helper's identity logged on every action; the student can revoke access.
- Guard against abuse: anomaly detection on one helper submitting many files, a visible "this service is free" notice, and a complaint link for overcharging.
- Shared phones are common. Do not tie identity to a single phone number; allow a second contact such as the school or CSC.

Disability

- UDID capture, slot handling for Divyangjan as the guideline requires, scribe or helper flag.
- No CAPTCHAs that block screen readers; use OTP and rate limits instead. Timeouts that extend on request.

Standards and testing

- GIGW 3.0 (WCAG 2.1 AA, mobile-first, aligned with the RPwD Act) is the Indian government guideline. Say "designed to align" until you have tested.
- Test with keyboard-only navigation, TalkBack and NVDA, a contrast checker, and an automated scan (axe or Lighthouse). If you can, test with a user who has a disability.
- Run a small usability test with three to five students in Hindi; report where they stalled.

Respect in design

- Avoid stereotyped imagery of tribal communities; use neutral illustrations and plain language.

- Ask permission before using any real person's photo; use synthetic data and illustrations in the demo.
- Name the communities' own words for things where officials confirm them, and avoid jargon like "beneficiary" in student-facing text.

12. Trust, privacy, security and human oversight

Judges from government backgrounds look for this section first. The rule is short: AI may flag, pre-fill, rank and explain. It never approves, rejects or selects.

Oversight mechanics

- Every decision logs officer id, time, the AI signals shown, and the mandatory reason.
- A rejection needs a second officer; an appeal goes to a different officer.
- Override rate and audit-sample miss rate are tracked and visible (section 8).
- Every flag shows its rule id, evidence box and one plain sentence, so a person can disagree with it.
- Each AI component has a short model card: what it does, inputs, known failures, confidence handling, and who can turn it off.
- Bias check: report flag rates by script (Hindi, English), by document type and by State, so a name format or a language never becomes a reason to be flagged more often.

Role-based access

Role	Can see	Can do	Cannot
Student	Own file and status	Edit before lock, respond to deficiency, appeal	See other students or AI risk scores
Helper	Files of consenting students, scoped	Fill, upload, submit on behalf	See files after consent is withdrawn
Institute officer	Own institute's files	Verify, return, escalate	See other institutes; reject without second officer
District or State officer	Files in their jurisdiction	Verify, return, reassign	Edit rules

Role	Can see	Can do	Cannot
Ministry officer	Central-scheme files	Final verification	Alter audit logs
Committee	Ranked pool with reasons	Rank, waitlist, sign off	See documents beyond need
Scheme owner	Aggregates, rules	Draft rules, simulate	Approve own rule change
Auditor	Read-only decisions and logs	Export evidence	Change anything

Implement as database row-level security, not only as screen hiding.

Privacy and law

- **DPDP Act 2023.** The Rules were notified on 14 Nov 2025; most notice, consent, breach and security duties take full effect around May 2027 (sources differ by a day). Verify dates.
- **Children:** under-18s are children under the Act and need verifiable parental consent. Pre-Matric covers classes IX and X, so build the guardian step. Persons with disabilities get similar protection.
- **Legitimate uses:** the Act has a route that may cover State benefits, and the Rules have exemptions. Say you would take legal advice rather than assume.
- **Aadhaar:** show and store only the last four digits and a reference token. Do not store full numbers or biometrics. Check UIDAI rules before any real integration.
- **Principles to show:** data minimisation, purpose limitation, retention schedule, a consent ledger including assisted-mode actors, access and correction rights, and a grievance contact.
- **Hosting:** a real deployment of this kind would sit on government-approved cloud in India. Name that as the production path even if your prototype uses a commercial region.

Security controls

- Officer multi-factor sign-in, short sessions, and device checks for admin roles.
- Private storage with short-lived signed links; encrypted documents and separate key management; virus scan and file-type validation on upload.

- Store extracted fields and a document hash rather than reprocessing originals, and purge originals on the retention schedule.
- Append-only, hash-chained audit log so tampering shows.
- Rate limits and abuse controls on OTP, upload and status endpoints; protection against bots on public pages.
- Treat OCR text as untrusted input (prompt-injection resistance, section 7).
- Backups, disaster recovery plan, and a breach response runbook. India's CERT-In rules set short incident-reporting windows and log-retention periods; verify the current terms.
- Independent security test before any real deployment.

Failure modes and fallbacks

What fails	What the system does
OCR service down	Files route to manual entry mode; no silent pass
Language model down	Deterministic template explanations
Rule engine error	File stays in queue with an error flag; no auto decision
Message gateway fails	Retry, then fall back to another channel and alert the helper
Payment API times out	Idempotent retry; no double payment
Duplicate or tamper signal	Flag and prioritise; the officer decides

Insider and helper abuse

- Conflict-of-interest recusal, anomaly detection on approval patterns by officer, second-officer rule on rejections, and sampling of auto-clean files.
- Limits and alerts on helpers who submit unusually many files, and audit of their actions.

13. Architecture, data model and tech stack

One database, one queue, small workers and swappable adapters keep the prototype buildable by a small team while the diagram still reads as production-grade.

Suggested stack

Layer	Choice	Why
Front end	Next.js PWA, Hindi and English message catalogues, accessible component set	Mobile-first, offline drafts, one codebase for all portals
Database and auth	Supabase Postgres with row-level security, Storage, Realtime	Per-role access in the database, live status updates
Queue and jobs	Postgres queue (pgmq) with scheduled jobs, or a hosted job runner	Fewer moving parts; idempotent retries
Rule engine	Pure TypeScript package, JSON schema validation, unit and property tests	Deterministic and testable, no AI inside
Document pipeline	Worker that calls the primary OCR, falls back to a second engine on low confidence	Matches section 7
Language model	Used only to explain flags and summarise files; template fallback	Never decides
Notifications	Adapter with sandbox or mock SMS and WhatsApp	Real templates, no real sends in the demo
Documents out	PDF generation for receipts, decision sheets, rank sheets	Audit and committee needs
Charts	Chart library for dashboards	Quick, consistent
Observability	Error tracking, structured logs, timing per pipeline stage	Lets you quote real latency numbers

Core data model

Entity	Key fields
scheme, scheme_version	owner, year, JSON rules, status (draft, approved, live), approver

Entity	Key fields
applicant_profile	registration id (hashed), name, DOB, State, contact, language, consent flags
application	applicant, scheme_version, state, submitted_at, locked_at
document	application, type, storage path, hash, quality score
extraction	document, field, value, confidence, bounding box, engine
flag	application, rule id, severity, evidence reference, status
deficiency	application, flag, message, due date, resolved_at
decision	application, officer, stage, outcome, reason code, signals shown (snapshot), time
selection, rank_entry	scheme_version, score components, slot, category, tie-break note
payment	application, batch, status, reason code, retry count
renewal, progress_report	application, year, status
institute	code, State, notified years, trust score
notification	recipient, channel, template, language, delivery status
consent_ledger	subject, actor (including helper), purpose, time, revoked_at
grievance, appeal	application, officer, stage, resolution
audit_log	actor, action, object, previous hash, hash

Adapters: build once, label as mock

Each adapter has one interface and one response shape that matches the real service. A flag switches mock to real. The UI shows a "MOCK" badge wherever a mock answers. Real integrations need government onboarding, so say that plainly.

Adapter	Mock behaviour	Real service later
Identity and One Time Registration	Returns a fake registration id and masked Aadhaar	Aadhaar eKYC and face authentication through the scholarship portal
DigiLocker	Returns signed sample certificates for a few test users	DigiLocker issued-document API
Bank seeding check	Returns seeded or not seeded with real-style reason text	NPCI mapper
Payment	Accepts a batch, returns success or failure codes	Public financial management system
SMS and WhatsApp	Writes to a log and shows a phone-style preview	Registered SMS templates; WhatsApp business API
Language services	Cached translations from vetted templates	Bhashini, Adi Vaani
Institute and student master data	Seeded table	AISHE code, UDISE+, academic bank ID

Scale and performance notes

- Deadline-day load is the real stress. Make every pipeline job idempotent, queue uploads, and report per-stage timing.
- Run a load test that simulates the last-day spike and show one chart of queue depth and completion time.
- Estimate cost per application as pages times OCR price plus model tokens. Measure it in the prototype and quote your own number.
- Index by scheme, year, institute and state; partition large tables by scheme year when you describe the production path.

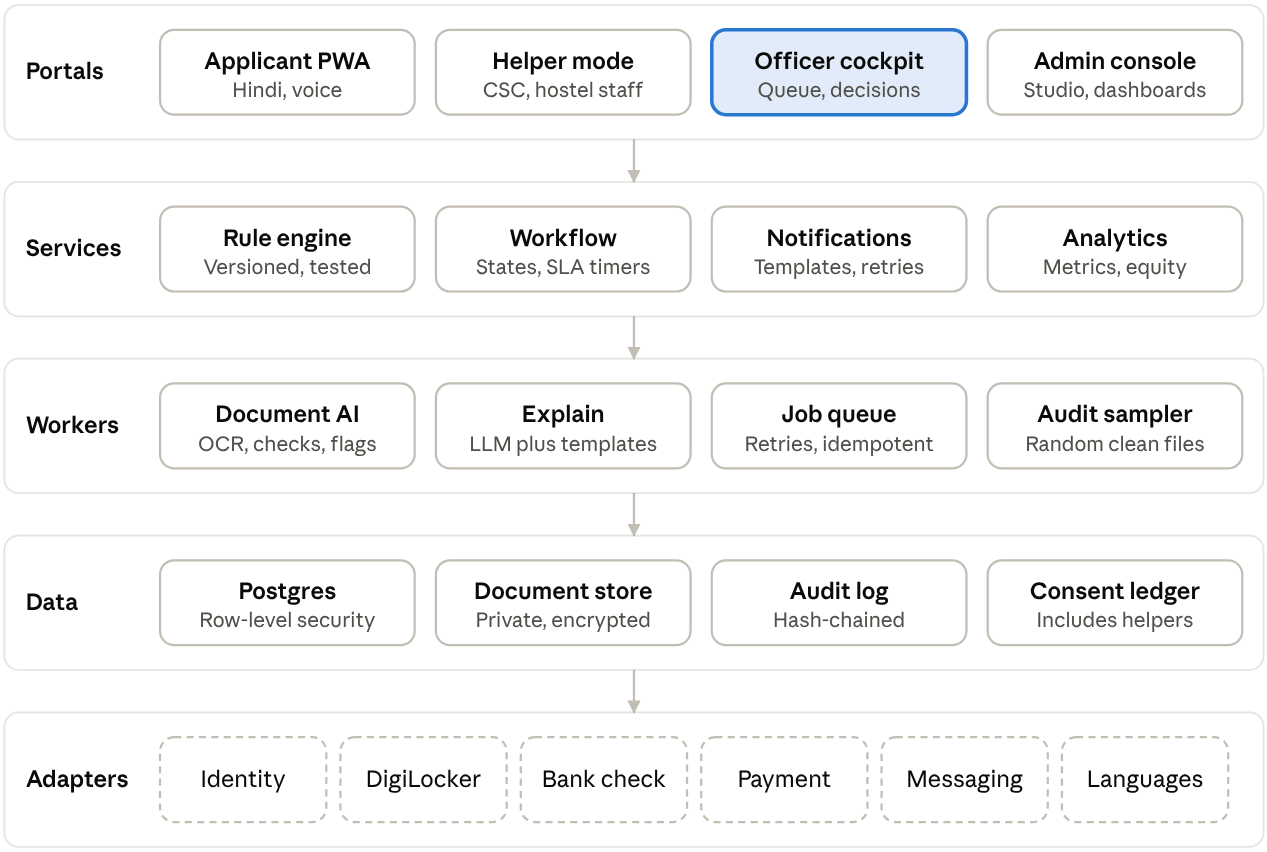
Demo reliability

- Cache the OCR output for the demo documents and keep a live-versus-cached switch.
- One seed script and a reset button to restore the demo state.
- Record a backup screen capture of the full flow in case the network fails.

Architecture drawing

The drawing below shows the portals, services, workers and adapters and where each human gate sits.

Five layers on one database; every outside service sits behind an adapter



Dashed boxes are mocks that return the real response shape; a flag swaps in the real service.
The highlighted box is the surface where a human decides.

architecture · 5 layers, 6 adapters

Portals call services, services queue work for the workers, everything lands in one database, and each outside system is reached only through an adapter.

14. Implementation plan

The plan assumes a 48-hour build window and a team of three to four. Stretch or shrink the hour marks in proportion if the event is shorter or longer. The order matters more than the hours: the golden path must work end to end before anything else is added.

The golden path (non-negotiable)

Official edits a rule in Rule Studio, student applies in Hindi, pre-flight catches a problem, OCR flags a mismatch, the student fixes it, an officer verifies with a reason, the committee ranks, the student sees a WhatsApp-style status, and the dashboard shows pendency dropping. Every phase below serves this line.

Phases

Phase	Window	Build	Done when
0. Prep before the clock	Before the event	Verify guideline values; write the scheme config for the main scheme; make 40 to 60 synthetic documents with ground truth; collect reason-code lists; draft message templates; sketch screens; write shared type definitions and the master prompts for each module. Check the event rules on pre-built work first.	Rule config reviewed against the guideline PDF; synthetic set ready; screen list agreed
1. Foundations	Hours 0 to 4	Project, auth, roles, row-level security, scheme config tables, seed data, Rule Studio v0 (edit values, publish a version), audit log trigger	Three roles can sign in and see only their data; a rule edit creates a new version
2. Application and upload	Hours 4 to 12	Applicant form (Hindi and English), draft save, upload with quality gate, mocked identity step, pre-flight check skeleton	A student submits a complete file with documents attached
3. Document intelligence	Hours 12 to 22	Classify, OCR with fallback, extraction, validators, cross-checks, flags with evidence boxes, confidence routing	Injected faults are caught on the synthetic set; field accuracy measured

Phase	Window	Build	Done when
4. Officer cockpit	Hours 22 to 30	Risk-ordered queue, side-by-side view, three decisions with reason codes, second-officer rule, audit sampling, decision sheet	An officer clears a mixed batch; every decision logged with signals shown
5. Applicant loop	Hours 30 to 36	Status timeline, deficiency copilot with exact-fix cards, re-upload, in-app and mock WhatsApp notifications	A flagged student fixes and re-submits without help
6. Selection, payment, admin	Hours 36 to 42	Merit ranking with slots, committee view, sign-off, payment mock with reason codes and seeding doctor, dashboard with five headline metrics	Ranked list published; a failed payment shows a plain-language fix
7. Second scheme and simulator	Hours 42 to 46	Post-Matric ST as config only; what-if simulator in Rule Studio	Second scheme runs with zero code change
8. Polish and rehearsal	Hours 46 to 48	Demo data reset, backup recording, README, pitch deck, three full rehearsals	Demo runs in five minutes twice in a row

Team split (four people; merge if fewer)

- **Applicant side:** form, uploads, status timeline, language, accessibility.
- **Pipeline:** rule engine, OCR, validators, flags, notifications adapters.
- **Officer and admin:** cockpit, decisions, selection, dashboards.
- **Data, demo and pitch:** synthetic documents, evaluation numbers, slides, rehearsal, domain accuracy.

If you build with AI coding tools, give each module a contract first (types and JSON schemas shared across the repo), write rule engine tests before the screens, and keep one person responsible for integration at every phase boundary.

Scope guard

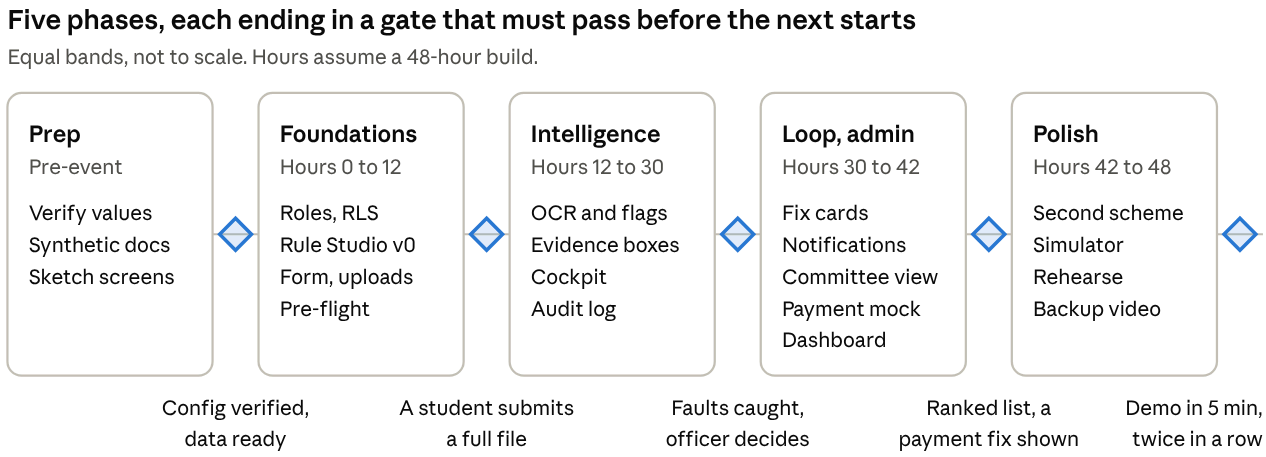
- Anything not on the golden path needs a trade: drop something of equal size.
- Cut in this order if time runs out: dropout early warning, renewal autopilot, voice guidance, extra languages, duplicate graph visuals, tamper heuristics beyond two or three.
- Never cut: human gates, audit log, pre-flight check, evidence boxes, the second-scheme config proof, measured accuracy numbers.

Multi-round events

If the event has an idea round and a build round, keep the same story in both. The idea round needs sections 2, 3, 7 and 12 as slides; the build round needs the golden path; the final needs the numbers.

Drawn roadmap

The drawing below shows the same phases with the gates between them.



roadmap · 5 phases, 5 gates

Prep is phase 0; Foundations covers phases 1 and 2; Intelligence covers 3 and 4; Loop, admin covers 5 and 6; Polish covers 7 and 8. A diamond is a gate: the next band does not start until it passes.

15. Demo script, pitch and judging strategy

Top-level juries see many portals. They remember a clear story with one human moment, a live failure that gets caught, and numbers you measured yourself.

Five-minute demo script

1. **Rule Studio (30 seconds):** a scheme owner raises the income ceiling and publishes version 2. The simulator shows how many synthetic applicants change status.
2. **Student applies in Hindi (60 seconds):** voice-guided form; the pre-flight check catches a bank seeding gap and says in plain words how to fix it.
3. **Document intelligence (60 seconds):** upload shows an OCR read with evidence boxes. It flags a name mismatch and an expired income certificate with rule ids and confidence.
4. **Deficiency copilot (30 seconds):** the student gets an exact-fix card in Hindi, with a read-aloud button, and re-uploads in place.
5. **Officer cockpit (60 seconds):** a risk-ordered queue; the officer approves a clean file with a reason and escalates a duplicate account. The decision sheet shows the signals that were visible.
6. **Committee (30 seconds):** ranked list with explanations and the equity preview before publishing.
7. **Close (30 seconds):** a WhatsApp-style status message, then the dashboard with pendency falling and the five headline metrics.

Show one failure on purpose (a forged-looking certificate that the system flags only as a risk) and say what it cannot catch. Honesty reads as maturity.

Pitch structure (ten slides)

1. Problem in numbers: three sourced figures from section 3.
2. A person: the Gond PhD applicant and the nodal officer.
3. Principle: AI flags, officer decides.
4. Live demo.
5. How it works: the flow drawing and the scheme config.
6. Evidence: field accuracy, flag precision and recall, time-to-verify test.
7. Inclusion: languages, voice, assisted mode, accessibility tests.
8. Trust: oversight mechanics, privacy, security, mocks labelled.
9. Path to real use: pilot at one institute during a live cycle, then State rollout; Ministry precedent of adopting student work from the Smart India Hackathon (verify).
10. Ask and next steps.

What judges usually weigh, and your proof

Likely criterion	Your proof
Impact and relevance	Sourced audit failures tied to features; tribal-first scope
Innovation	Pre-flight check, deficiency copilot, Rule Studio simulator, audit sampling
Technical depth	Tested rule engine, OCR with fallback and calibrated confidence, hash-chained audit log
Feasibility	Config-only second scheme, adapters with real response shapes, honest mock labels
Scalability	Queue design, load test on deadline spike, scheme packs
Design and accessibility	Hindi-first, voice, low-bandwidth, tested against GIGW checks
Presentation	One story, one failure shown live, numbers on slides with sources

Position against the existing portal

Do not say you replace the national scholarship portal. Say you add a verification, tracking and communication layer that integrates with it through the same identity and payment rails. Criticising what exists makes the room defensive; showing the gap your tool closes does not.

Hard questions to prepare

Question	Answer in one breath
What if the OCR is wrong?	It is a signal with a confidence score; the student confirms what it read, and the officer sees the evidence box and decides. We measured accuracy and show the misses.
Does the AI decide tribe status?	No. States issue and verify certificates. We check consistency and route doubts to the issuing authority.

Question	Answer in one breath
How do you protect Aadhaar and personal data?	Last four digits only, encrypted documents, row-level access, consent ledger, purpose-limited retention, and legal review under the data protection law before launch.
Is the Aadhaar and payment integration real?	No, they are mock adapters with the real response shapes, labelled on screen. Real integrations need government onboarding.
How do you catch forged documents?	Risk signals plus issuer verification plus random audits. No single method catches a competent forger, and we say so.
Why would officers trust it?	They can see why every flag fired, they decide every case, and we measure and publish the override rate and audit misses.
Can it scale to lakhs of students?	Queue-based and idempotent; we ran a deadline-spike test. Production would sit on government-approved cloud.
What about languages we have not tested?	We claim only what a speaker has checked, and use human-vetted templates for notices.
Who maintains the rules?	Scheme owners, through Rule Studio with maker-checker approval and full version history.

16. Risks, what to mock, and what to verify

The main risks are scope and domain depth, not technology. The fix for both is a small golden path built on checked facts.

Risk register

Risk	Why it matters	Mitigation
Scope creep across six stages plus analytics	Half-built stages lose to one finished path	Golden path, trade rule, cut order (section 14)

Risk	Why it matters	Mitigation
Shallow domain detail	Judges with scheme experience spot it fast	Verify guideline values; speak to a nodal officer or CSC operator; use real reason codes
Uneven Indic OCR	Hindi accuracy did not track English strength in a June 2026 preprint	Test on own scans, show confidence, keep human correction, report failures
Real systems need onboarding	Aadhaar, DigiLocker, payment rails cannot be called freely	Mock adapters with real response shapes, labelled
Tribal identity handled badly	State-specific lists, disputed cases, and a migrant can claim ST only in the home State	AI never decides; route to issuing authority
Unsourced statistics	One wrong number damages trust	Quote audit reports and parliament replies; avoid blog figures
Overclaiming on law or accessibility	Legal and standards claims are checkable	Say "designed to align" until tested; take legal advice
Demo failure (network, quota, latency)	A broken live demo costs more than a modest one	Cached OCR, reset script, backup recording
Cost or rate limits on OCR and model APIs	Surprise failure mid-demo	Cache results, fallback engine, test quota in advance
Team coordination and attention split	Other events and projects compete for time	Fixed roles, integration checkpoints, written contracts
"AI decides" perception	Undermines the oversight promise	Say and show the human gates in every screen
Vendor accuracy claims	Marketing numbers differ from your documents	Report your own measured accuracy

Real versus mocked

Real in the prototype	Mocked and labelled
Roles, workflow and state machine	Aadhaar eKYC and face authentication
Rule engine with versions and tests	DigiLocker
Audit log with hash chain	Bank seeding check and payment rail (use real reason codes)
OCR on synthetic documents with fallback	SMS and WhatsApp gateways
Basic tamper heuristics	Language services beyond cached vetted templates
In-app notifications and status timeline	Institute master data beyond a seeded list

Use synthetic data only. Never put real student documents or real Aadhaar numbers into the prototype or a demo.

Facts to verify before they reach a slide

The live cycle makes some facts time-sensitive. On 29 Oct 2026, the institute verification date of 15 Nov 2026 is 17 days away and the Ministry date of 30 Nov 2026 is 32 days away, per the pack.

Fact	Where it came from	Verify at
Cycle dates for A.Y. 2026-27 (letter of 23 Jun 2026; 15 Nov and 30 Nov deadlines)	Search excerpt of the Ministry letter	The letter PDF itself
Income ceilings (Rs 6 lakh scholarship; Rs 2.5 lakh pre-matric) and notified-institute count (265)	Guidelines 2021-26 and 2026 list excerpts	Current guideline PDF and notified list
Fellowship and overseas award counts (about 750 and about 20)	Third-party explainers	Official guidelines
Verification chain and payment flow	NSP SOP (July 2019) and demo pages	Latest SOP and NSP help pages

Fact	Where it came from	Verify at
About 30 lakh ST students covered by DBT	Ministry page excerpt	Ministry website or a parliament reply
Audit figures (Jharkhand, Odisha, Himachal, Tamil Nadu, minority scholarships)	Audit reports and news coverage	The audit reports on the auditor's site
Payment failure notices (Tripura, Arunachal, Karnataka)	State notices and news	The notice or article itself
Hindi OCR accuracy numbers (58.5, 40.5, 75.2 chrF++)	June 2026 preprint	The preprint's tables
Adi Vaani languages and the Gondi accuracy remark	News coverage	Ministry or developer announcement
Data protection Rules dates (notified 14 Nov 2025; duties around May 2027)	Third-party summaries; sources differ by a day	The Gazette notification
Ministry co-developing a platform with Smart India Hackathon finalists	Parliament reply excerpt	The reply itself
Reservation terms in fellowship guidelines (including Divyangjan slots)	Excerpt	Current guideline PDF

Keep a source log with the link, the date checked and the initials of whoever checked it.

17. Master checklist: standout ideas and small details

Use this as the "did I miss anything" list. The ranking table orders the standout ideas by how much they add to the demo per unit of work.

Standout ideas, ranked

#	Idea	Effort	Demo impact	Priority
1	Pre-flight "will this pass?" check	Medium	High	Must

#	Idea	Effort	Demo impact	Priority
2	Deficiency copilot (voice, local language, guided photo)	Medium	High	Must
3	Rule Studio with what-if simulator	Medium	High	Must
4	Officer cockpit with audit sampling	Medium	High	Must
5	Seeding doctor (payment reason code to next steps)	Small	High	Should
6	WhatsApp and SMS status bot	Small with mocks	Medium	Should
7	Institute trust score	Small	Medium	Should
8	Duplicate account graph	Medium	Medium	Should
9	One-page decision sheet with tamper-evident log	Small	Medium	Should
10	Synthetic data generator (also your evaluation set)	Medium	Medium	Must for proof
11	Renewal autopilot	Small	Medium	Could
12	Dropout early warning	Small	Low	Could

Extra ideas beyond the research pack

- **Verify-this-letter page:** a QR code on every sanction letter and receipt opens a public page that confirms it is genuine. This answers fake-letter problems cheaply.
- **Enrolment roll cross-check:** institutes upload a signed enrolment list; applicants missing from it are flagged. This directly targets the ghost-beneficiary findings.
- **Document wallet:** upload once, reuse across schemes and years, with expiry reminders.
- **"Which scheme fits me" comparison** in the public eligibility checker.
- **Officer training mode:** a sandbox with synthetic files so officers can practise before go-live.
- **Open API and webhooks** so State portals and institute systems can plug in.
- **Public transparency page** with aggregate numbers only.

- **Scholarship passport:** a student's multi-year timeline of applications, renewals and payments.
- **Help desk loop:** a call-back request that opens a ticket for the assisted-mode helper.

Checklist

Product

- [ ] Public eligibility checker with no login
- [ ] Hindi and English across every screen; voice read-aloud on the applicant flow
- [ ] Draft save, offline drafts, resumable uploads
- [ ] Pre-flight check with at least five checks (seeding, name, certificate date, quality, missing document)
- [ ] Student confirmation of OCR-read values
- [ ] Receipt with application ID and QR
- [ ] Status timeline with plain-language reasons and expected wait
- [ ] Deficiency cards with sample photos and deadline countdown
- [ ] Appeal and grievance paths
- [ ] Officer queue by risk and age, side-by-side view, shortcuts
- [ ] Coded rejection reasons in each language
- [ ] Waitlist with auto-promotion
- [ ] Payment status with seeding doctor
- [ ] Renewal flow that asks only for current-year proof

Data and AI

- [ ] Versioned scheme config for two schemes
- [ ] Rule engine unit tests and golden cases
- [ ] 40 to 60 synthetic documents with injected faults and ground truth
- [ ] Measured field accuracy and flag precision and recall
- [ ] Confidence thresholds tuned and documented
- [ ] Grounding check on every model-extracted value
- [ ] Model card for each AI component; bias check by script and State

Trust and operations

- [ ] Row-level security per role, tested
- [ ] Hash-chained audit log with officer id, time, signals shown, reason

- [ ] Second-officer rule on rejection; different officer on appeal
- [ ] Random audit sample with blind review option
- [ ] Consent ledger including assisted-mode actors
- [ ] Aadhaar last four digits only; retention schedule written
- [ ] Failure fallbacks demonstrated (OCR down, model down)
- [ ] Load test of the deadline-day spike

Pitch

- [ ] Three sourced numbers on the problem slide
- [ ] Every mock labelled on screen and on a slide
- [ ] One live failure shown and explained
- [ ] Backup recording, reset script, three rehearsals
- [ ] Answers rehearsed for the nine hard questions in section 15

18. Open questions and pre-event research tasks

Several facts change the plan. Answer the open questions first, then work through the task list in priority order. Stop researching once the "verify" table in section 16 is clear; further reading has diminishing returns compared with a rehearsed demo.

Open questions

Question	Why it matters	How to settle it
Exact owner and ID of this problem statement (assumed Ministry of Tribal Affairs from the wording)	Decides the domain language and which official sources to cite	Event organisers' brief or the problem statement sheet
Event stages, dates and submission format (deck, repo, video, live demo)	Sets the build window and which sections become slides	Event page or organisers
Are pre-built assets allowed (templates, datasets, code)?	Phase 0 depends on it	Event rules; ask organisers
Published judging criteria and weights	Re-weights the demo script and slides	Event page

Question	Why it matters	How to settle it
Team size and roles	Changes the split in section 14	Team decision
Primary scheme: National Fellowship and Scholarship or Post-Matric ST	Drives the rule config and sample documents	Pick National Fellowship and Scholarship unless organisers say otherwise
Home-State angle (Madhya Pradesh certificates, State portal, local languages)	Local judges respond to local detail	Check which State documents and portals you can model from public samples
Access to a nodal officer or CSC operator for feedback	A single conversation sharpens the cockpit	Ask your college scholarship cell or a local CSC
Demo languages beyond Hindi and English	Tribal-language claims need a speaker	Find a speaker, or limit the claim to audio notices in tested languages
Venue internet and API quotas	Demo reliability	Test on the venue network; cache results

Pre-event research and prep tasks

Priority	Task	Output
1	Read the Ministry's scholarship and fellowship guidelines (2021-26)	Rule list with page references; filled-in scheme config
1	Read the 23 Jun 2026 letter, the One Time Registration FAQ and the verification SOP	Process map and deadline table
1	Open the audit reports behind each cited figure	Page-numbered source log
1	Check hackathon rules on pre-built work and submission needs	Written answer
2	Talk to one CSC operator, one scholarship clerk, two students	One page of quotes and top five failure points

Priority	Task	Output
2	Collect payment failure reasons and State notices (Tripura, Arunachal)	Reason-code table with plain-language fixes
2	Test two or three OCR engines on 20 of your own Hindi and English scans, printed and handwritten	Baseline accuracy table
2	Build the synthetic document generator and ground-truth labels	40 to 60 documents with injected faults
2	Draft notices in Hindi and have a native speaker check them	Approved template set
3	Gather public sample formats of State ST and income certificates	Fictional templates modelled on real layouts
3	Read the data protection Rules summary and the Gazette text	One page of design obligations
3	Prepare accessibility test checklist and tools	Test plan
3	Write demo seed script, reset button and backup recording plan	Demo kit
3	Schedule three timed rehearsals with a question round	Calendar entries

Sources used in the research pack

Official and primary: Ministry of Tribal Affairs scholarship page; Ministry guidelines 2021-26; Ministry letter of 23 Jun 2026; national scholarship portal SOP (July 2019), demo and One Time Registration FAQ; audit reports for Jharkhand and Kerala; Tripura tribal welfare notices; Himachal DigiLocker linking letter; Lok Sabha replies on ST certificates and the Forest Rights Act platform; GIGW 3.0.

News and other: audit coverage for Odisha, minority scholarships (Dec 2025), Tamil Nadu, Himachal, Arunachal and Karnataka; reporting on West Bengal ST certificates; Sarvam product pages; a June 2026 Hindi OCR preprint; Indic TrOCR repository; Adi Vaani coverage; third-party data protection summaries; third-party tampering guides. Links are in the research pack you shared; open each original before quoting.